Autonomous Delivery Robot

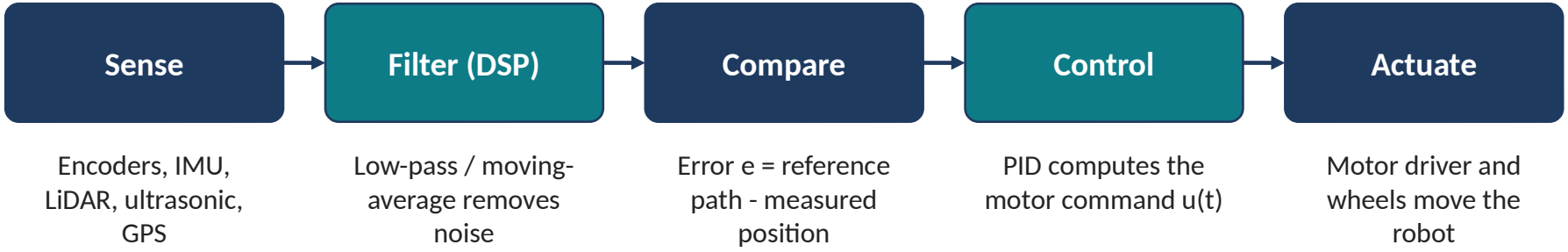
Feedback Control System and DSP-based Sensor Filtering

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1. Introduction and Objective

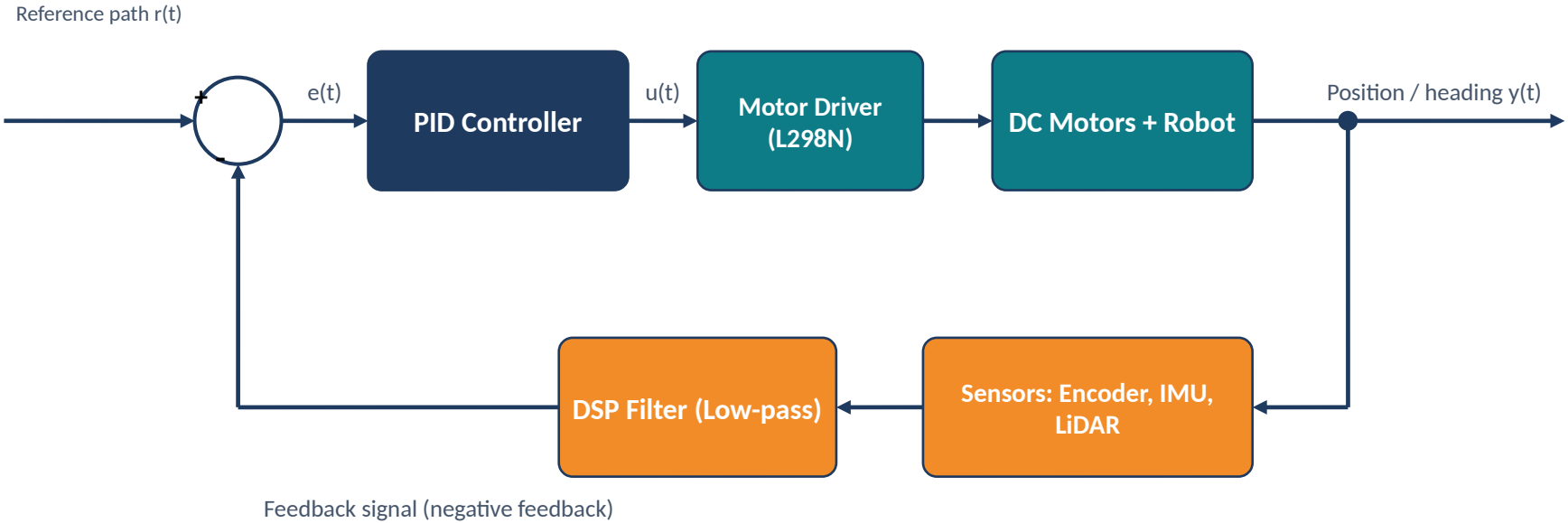
- An autonomous delivery robot carries parcels or food from a source to a destination without a human driver.
- Control: a path-tracking feedback controller steers the robot along the planned route.
- DSP: filtering of LiDAR/camera signals removes noise and supports reliable obstacle detection.
- Objective: keep the tracking error small and avoid obstacles using closed-loop control plus digital filtering.
- Applications: campus, hospital, restaurant and last-mile delivery.

2. Working Principle



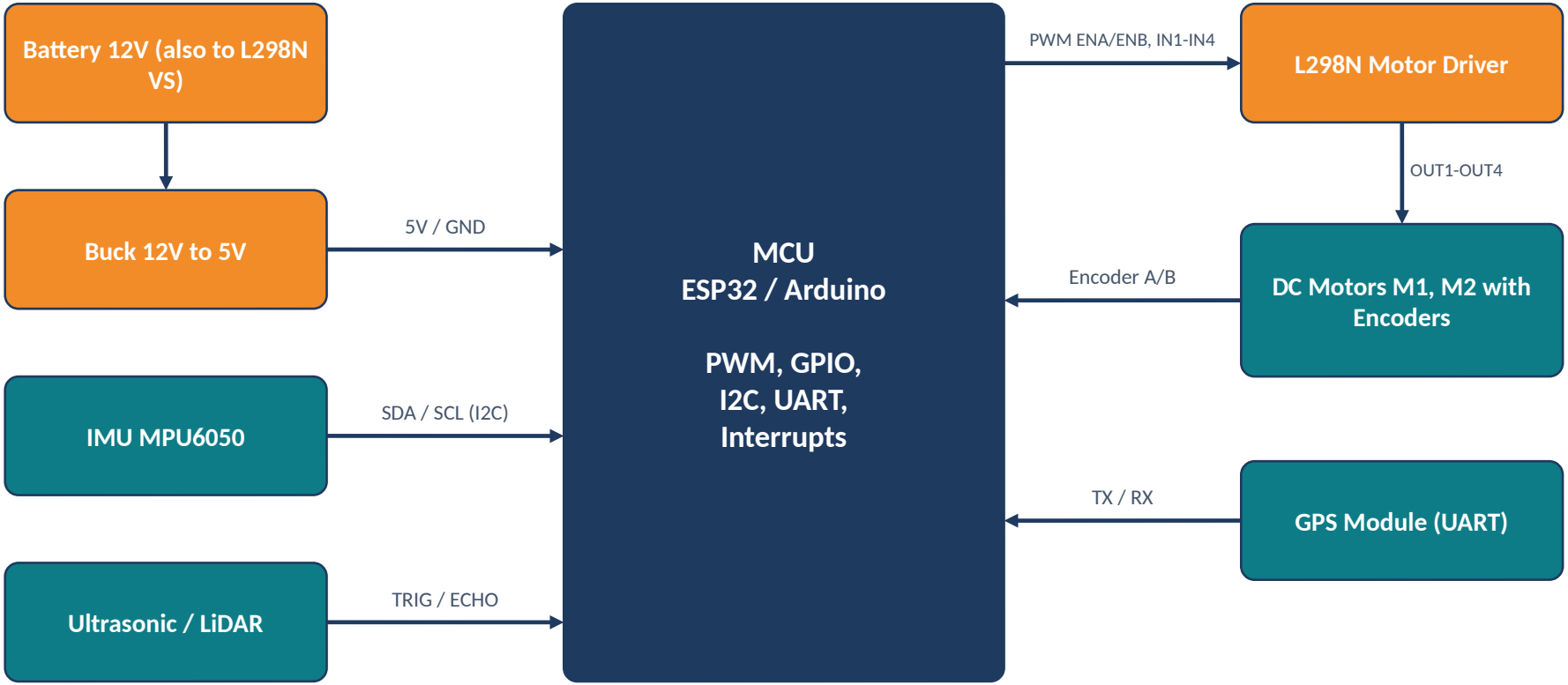
The loop repeats every few milliseconds. The measured output is fed back, so the robot continuously corrects drift, slip and disturbances.

3. Block Diagram (Closed-Loop Control)



Disturbances (slope, wheel slip, obstacles) act on the robot block. Orange blocks form the feedback path.

4. Circuit Diagram



5. Use of Feedback in Control System

Error signal
e(t) = r(t) - y_f(t)

PID control law
u(t) = Kp e + Ki integral(e dt) + Kd de/dt

Plant (motor + robot)
G(s) = 2 / [s(0.5s + 1)]

Closed loop (unity feedback)
T(s) = C(s)G(s) / [1 + C(s)G(s)]

- Negative feedback compares the measured output with the reference.
- P reduces the error, I removes steady-state error, D damps overshoot.
- Feedback rejects disturbances such as slip and slopes.
- It makes the robot track the path even if the plant model is imperfect.
- Chosen gains: Kp = 4, Ki = 0.5, Kd = 0.8 (stable closed loop).

6. DSP Filtering of Sensor Signals

- LiDAR/camera/encoder signals contain high-frequency noise.
- A digital 4th-order Butterworth low-pass filter (cut-off 5 Hz, fs = 100 Hz) is applied.
- filtfilt applies it forward and backward, giving zero phase delay.
- Simple alternative: moving average y[n] = (1/N) sum of x[n-k], k = 0..N-1.
- Clean distance values give reliable obstacle detection and a smoother feedback signal.

Signal chain

Raw sensor x[n]

v

Low-pass filter H(z)

v

Filtered y[n]

v

Summing junction / controller

7. MATLAB Code

```
clc; clear; close all;
%% 1. Plant: heading dynamics (motor + wheels)
s = tf('s');
G = 2/(s*(0.5*s+1));
%% 2. PID controller with unity negative feedback
C = pid(4, 0.5, 0.8);
T = feedback(C*G, 1);
[y,t] = step(T, 10);
disp(stepinfo(T));
%% 3. DSP: noisy LiDAR distance and filtering
fs = 100; n = 0:1/fs:5;
d  = 2 + 0.3*sin(2*pi*0.5*n);   % true distance
dn = d + 0.2*randn(size(n));    % noisy signal
[b,a] = butter(4, 5/(fs/2));    % 5 Hz low-pass
df = filtfilt(b,a,dn);
%% 4. Plots
subplot(2,1,1); plot(t,y,t,ones(size(t)),'--'); grid on;
title('Closed-loop step response');
subplot(2,1,2); plot(n,dn,n,df,'LineWidth',1.5); grid on;
legend('Noisy','Filtered'); title('DSP filtering');
```

8. Explanation of MATLAB Code

- tf('s') and G: transfer function of the motor-plus-robot heading dynamics (integrator with a 0.5 s lag).
- pid(4,0.5,0.8): creates the PID controller C(s) with Kp, Ki, Kd.
- feedback(C*G,1): closes the loop with unity negative feedback, giving T(s).
- step and stepinfo: simulate the response and report rise time, settling time and overshoot.
- butter(4, 5/(fs/2)): designs a 4th-order low-pass filter; cut-off is normalised by fs/2.
- randn adds noise to the true distance; filtfilt removes it with zero phase lag.
- subplot/plot: top graph shows tracking of the reference; bottom graph compares noisy and filtered LiDAR data.
- Needs Control System Toolbox and Signal Processing Toolbox.

9. Results and Conclusion

- Expected result: the step response settles at the reference (steady-state error is zero because of the integral action).
- The filtered LiDAR signal follows the true distance with the noise removed.
- Feedback control keeps the robot on its path; DSP filtering makes obstacle detection reliable.
- Future scope: sensor fusion (Kalman filter), SLAM, and GPS waypoint navigation.

Control (feedback loop) + DSP (filtering) = accurate and safe autonomous delivery.